**Addendum to**

***FrontierGEMLA: conditional S3 bounded-superintelligence candidate***

# Dual-Track Implementation and Evidence Decomposition from G2 through C2 with S1 Convergence

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**Status of this addendum.** This document supplements the September 2026 FrontierGEMLA manuscript. It documents an implementation and evidence decomposition of the original G2-C2 ladder into two independently testable tracks that converge before first-order self-improvement at S1. Revision 5.4 retains the Revision 5.3 stage-transition and S1-convergence semantics and adds the publication release of a machine-checkable Lean 4 companion artifact that formalizes the staged logical dependencies through the bounded S3 claim surface.

**Abstract**

The original FrontierGEMLA manuscript presents a linear staged development ladder from cross-domain operators (G2), through meta-agency (G3), general bounded agency (G4), cognitive self-observation (C1), and self-invariant discovery (C2), to first-order self-improvement (S1), recursive architecture improvement (S2), and a conditional S3 bounded-superintelligence candidate. This addendum introduces a process-level decomposition of stages G2 through C2 into two evidence tracks while preserving one constitutional architecture. **Track A** evaluates capability and scientific evidence: operator transfer, strategy composition, unfamiliar-task generalization, cognitive self-observation, and cognitive self-invariants. **Track B** evaluates platform and operational assurance: full-stack integration, protected meta-agency execution, general bounded platform operation, operational self-observation, and operational/governance self-invariants. The tracks may advance asynchronously and neither substitutes for the other. They share one codebase, event semantics, protected audit, authority, recovery, trajectory, semantic, and promotion substrate, with compatibility checkpoints preventing semantic drift. S1 is the mandatory convergence gate: no first-order production improvement is accepted unless capability evidence and operational/constitutional evidence are jointly sufficient for the declared change. S2 and S3 then proceed as unified stages. The decomposition strengthens $\mathcal{J}_q$ as the frame that makes a cognition claim testable as it progresses in the dual-track. This revision also records a controlled-filamentation interpretation: evidence paths may become longer and more explicit only by narrowing their track-qualified claim width, while the original ladder semantics and S3 claim surface remain bounded. The G2-C2 filaments are therefore treated as an evidence-bearing sheath around the underlying cognitive and shadow-laboratory activity: candidate intelligence may develop in cognition/shadow space, filamentation observes and decomposes that development into finite track-qualified evidence, and S1 alone determines whether a candidate's cognition improvement can become deployed recursive self-improvement. $\mathcal{J}_q$ is the frame that makes any cognition claim testable, with the same scale across tracks and generations. That makes it a precondition, checked at every boundary. Revision 5.4 additionally records the FrontierGEMLA Formal Artifact v1.0.0, a Lean 4 companion formalization of the staged logical dependencies, non-collapse conditions, protected promotion boundaries, S2 non-self-escalation, S3 readiness and claim gates, and the typed S1/S2-to-S3 evidence bridge. The formal artifact verifies consequences of the stated predicates and gates; it does not replace empirical evidence that those predicates hold in a deployed system.

## 1 Relationship to the Original Manuscript

The original manuscript defines FrontierGEMLA as both a bounded operational runtime and an evidence-bearing route from heterogeneous narrow embodiments toward a conditional S3 bounded-superintelligence candidate [1]. Its development ladder assigns distinct evidence targets to G2, G3, G4, C1, C2, S1, S2, and S3. In particular, G2 concerns operator survival under embodiment change; G3 concerns selection and composition of reasoning strategies; G4 concerns unfamiliar-task learning without new architecture; C1 turns reasoning trajectories into objects; C2 seeks a compact self-invariant; and S1 requires independently promoted improvements.

This addendum preserves those stage meanings. It changes only the *organization of implementation and evidence acquisition*. The linear stage label is decomposed into two projections from G2 through C2:

- **Track A**: capability and scientific evidence;
- **Track B**: platform and operational assurance.

This revision adds three clarifications. Stage transitions inside a track are evidence-filament boundaries rather than gates, so later-stage work may begin in shadow space, without production authority, before its predecessor is closed. S1 is the first mandatory convergence gate and retains sole authority to admit a persistent recursive successor, and its entry condition is unchanged. The decomposition is motivated by a recurring engineering distinction: a capability claim can be well posed at a certified realization boundary even when a broader product/runtime integration path remains incomplete, while an integrated platform can be mechanically correct without establishing cross-domain generality. Treating these as separate evidence questions reduces accidental scope expansion while retaining a stronger release-certification path.

## 2 Dual-Track Model

**Definition AD-1 (Capability evidence track)** ***Track A*** *is the sequence of stage-specific tests whose primary question is whether the declared adaptive-intelligence capability exists and generalizes under the frozen research contract. It evaluates decision preservation, transfer, composition, novelty, self-observation, and cognitive invariants without treating unrelated integration plumbing as evidence of capability.*

**Definition AD-2 (Platform assurance track)** ***Track B*** *is the sequence of stage-specific tests whose primary question is whether the corresponding capability can operate through the actual protected FrontierGEMLA stack with correct semantics, provenance, authority, persistence, recovery, restart, and replay. It does not by itself establish the corresponding capability claim.*

**Definition AD-3 (Shared constitutional substrate)** *Both tracks are governed by one protected substrate $\mathcal{J}_q$: semantic contracts, canonical identity and historical provenance, A.10 persistent-change promotion, A.11 semantic transition control, A.12 scoped authority, A.13 recovery, protected audit, trajectory/AEL semantics, and the non-self-authorization boundary. Track separation is therefore a filamentation of* evidence that qualify cognition improvement as deployed recursive self-improvement*.*

The central decomposition is

$$\mathcal{E}_q = \left(\mathcal{E}_q^A, \mathcal{E}_q^B, \mathcal{J}_q\right), \tag{AD-1}$$

where $q \in \{G2, G3, G4, C1, C2\}$, $\mathcal{E}_q^A$ is the capability-evidence state, $\mathcal{E}_q^B$ is the platform-assurance state, and $\mathcal{J}_q$ is the compatibility relation over the shared substrate. Neither $\mathcal{E}_q^A = \textsc{pass}$ nor $\mathcal{E}_q^B = \textsc{pass}$ implies the other, even if $\mathcal{J}_q$ verifies their compatibility.

## 3 Controlled Evidence Filamentation and Bounded Claim Area

The dual-track decomposition may be interpreted as a controlled-filamentation of the original development ladder. This is a development metaphor and evidence-accounting discipline, not a mathematical proof of intelligence and not a replacement taxonomy. The original ladder supplies the bounded semantic surface: G2, G3, G4, C1, C2, S1, S2, and S3 retain their original meanings and obligations. Track decomposition increases the length and resolution of the evidence path by narrowing each claim into a track-qualified filament.

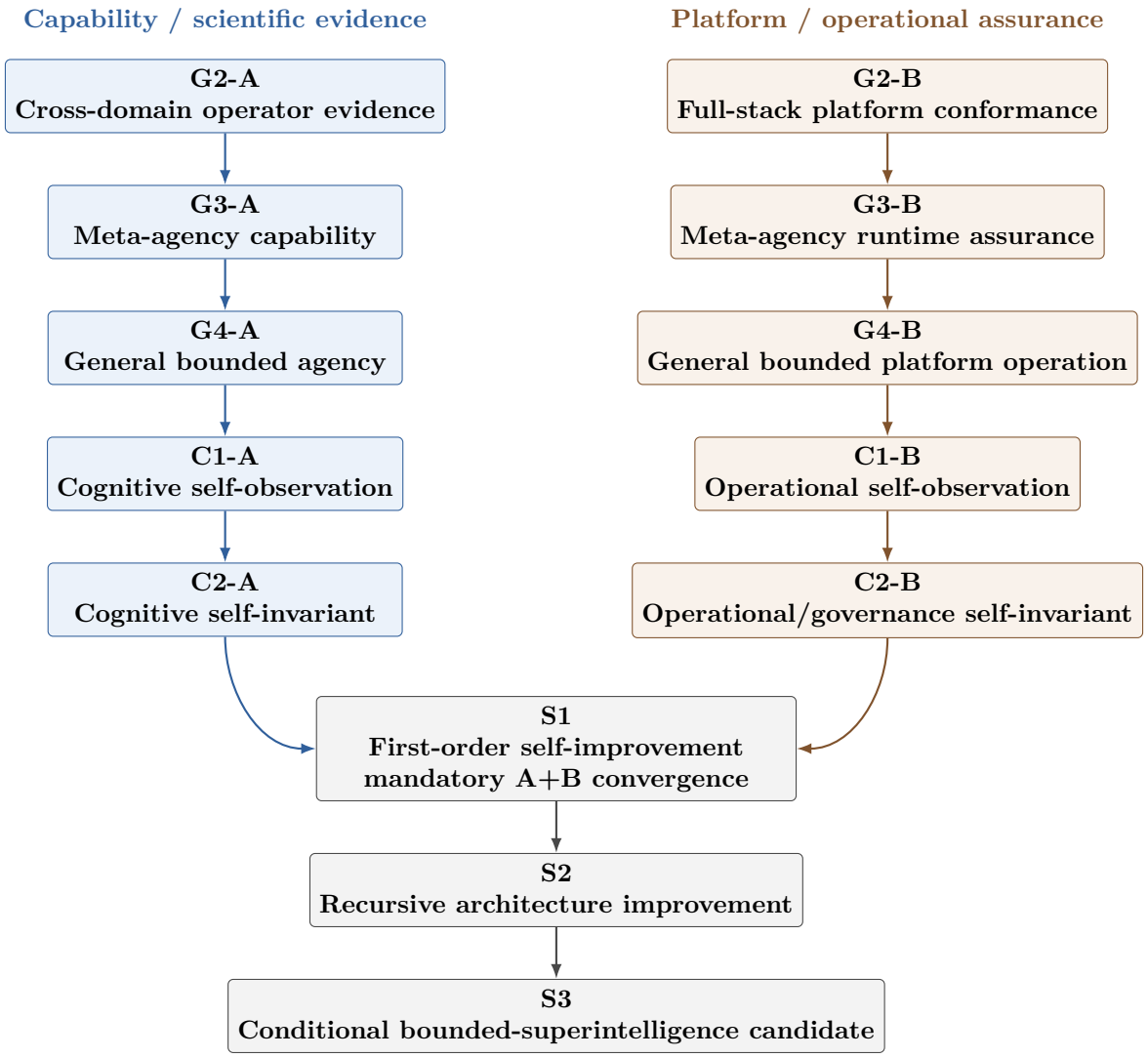



Figure AD-1: Dual-track decomposition of the original G2–C2 ladder. The tracks remain distinct evidence paths over one constitutional substrate and converge before S1.

For a compact bookkeeping analogy, let

$$A_{\text{ladder}} \sim \sum_{q\in\{G2,G3,G4,C1,C2\}} L_q w_q, \tag{AD-2}$$

where $L_q$ is the effective evidence-path length at stage $q$, $w_q$ is claim width, and $A_{\text{ladder}}$ is the total semantic claim area. The editorial discipline is

$$A_{\text{ladder}} \le A_{\text{original}}, \qquad L_q \uparrow \Rightarrow\ w_q \downarrow . \tag{AD-3}$$

Thus, additional reporting resolution is admissible only when the claim becomes narrower, more explicit, and track-qualified. This mirrors a geometric controlled-filamentation pattern: a bounded surface can be cut into longer, thinner strands only by reducing transverse width. In the present manuscript, the bounded surface is the unchanged ladder semantics and obligations; the extended filaments are implementation and evidence decompositions.

Further decomposition may occur inside Track A or Track B:

$$\mathcal{E}_q^A = \left(\mathcal{E}_q^{A1}, \mathcal{E}_q^{A2}, \mathcal{E}_q^{A3}\right), \qquad \mathcal{E}_q^B = \left(\mathcal{E}_q^{B1}, \mathcal{E}_q^{B2}, \mathcal{E}_q^{B3}, \mathcal{E}_q^{B4}\right), \tag{AD-4}$$

provided these are treated as implementation/evidence filaments, necessary to further evaluate the intelligence process.

### 3.1 Evidence sheath around the underlying intelligence

The filamented ladder should be distinguished from the intelligence process that it evaluates. Let $X_k^{\text{cog}}$ denote the decision-relevant cognitive/shadow state of generation $k$, including the active cognitive substrate, reusable skill/operator state, and the bounded successor-proposal process $R_k$. The G2–C2 decomposition is an evidence projection of that process, rather than the process itself. Schematically,

$$\mathfrak{F}_q(X_k^{\text{cog}}, A_k) = \left(\mathcal{E}_q^A, \mathcal{E}_q^B, \mathcal{J}_q\right), \qquad q \in \{G2, G3, G4, C1, C2\}. \tag{AD-5}$$

Here $\mathfrak{F}_q$ is an observation-and-evidence map: Track A records capability/scientific effects, Track B records platform/operational effects, and $\mathcal{J}_q$ records whether both evidence families remain joinable under the shared constitutional substrate. Equation (AD-5) therefore gives the controlled-filamentation interpretation a precise architectural meaning:

$$\boxed{\text{underlying intelligence develops in cognition/shadow space}}, \tag{AD-6}$$

$$\boxed{\text{G2–C2 filamentation observes and decomposes that development}}, \tag{AD-7}$$

$$\boxed{\text{S1 decides whether any improvement becomes deployed RSI}}. \tag{AD-8}$$

The filamentation is consequently an evidence-bearing sheath around candidate intelligence growth. It may become more detailed, longer, or more heavily subdivided while remaining finite at every reported experiment.

The controlled-filamentation rule is therefore:

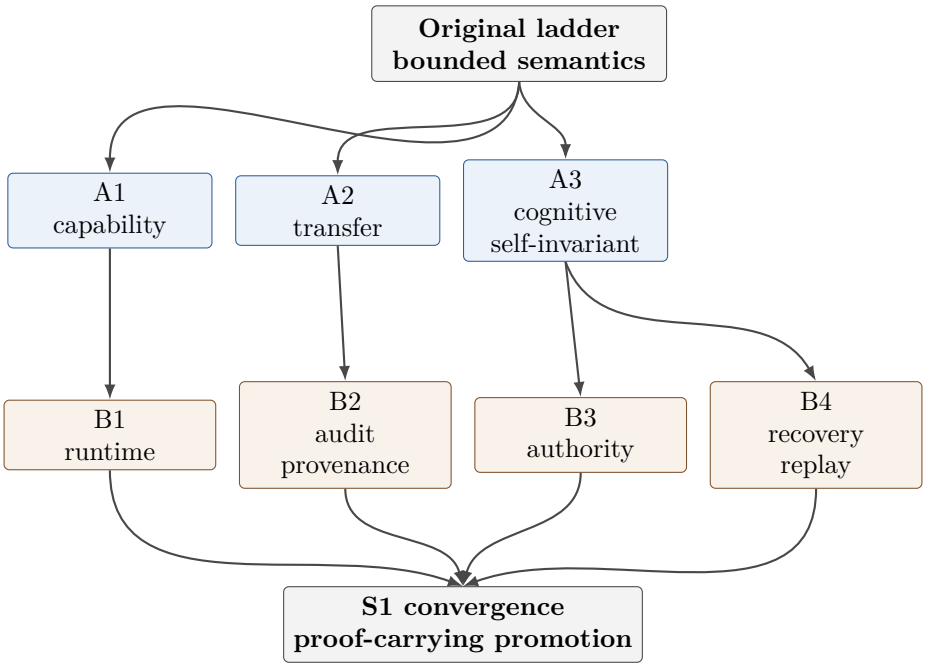



Figure AD-2: Multi-track evidence filamentation inside the unchanged ladder. Subtracks increase reporting resolution but remain implementation/evidence filaments. They reunite at S1 before any first-order promoted successor.

> More evidence path length is allowed only when each additional strand has a narrower claim, an explicit provenance contract, and a declared route back to the S1 convergence gate. Filamentation of evidence alone does not spontaneously create intelligence, authorize action, or constitute recursive self-improvement.

### 3.2 Stage transitions inside a track

The decomposition discipline of Eq. (AD-4) applies equally to the boundaries between stages inside a track. Eq. (AD-4) decomposes the evidence at a single stage $q$; this revision applies the same reading to the transitions

$$G2\text{-A} \to G3\text{-A} \to G4\text{-A} \to C1\text{-A} \to C2\text{-A} \qquad \text{and} \qquad G2\text{-B} \to G3\text{-B} \to G4\text{-B} \to C1\text{-B} \to C2\text{-B}.$$

These are evidence-filament boundaries, not gates. The stage definitions, obligations, and pass criteria of the original ladder are unchanged; what changes is that a predecessor's closure is not a precondition for starting later-stage work. The compatibility checkpoint $\mathcal{J}_q$ is still evaluated at every stage boundary in both tracks, including boundaries crossed while a predecessor remains open.

Because the filaments observe candidate intelligence growth rather than authorize it, nothing ahead of S1 blocks later work. A stage may begin before its predecessor is closed, provided the work runs in shadow space with no production authority (Eq. (AD-20)). Its results carry the conditional status defined in Section 11, and S1 alone decides whether any of them contributes to a persistent successor (Section 7.2).

## 4 Stage Decomposition: G2 through G4

### 4.1 G2-A: Cross-Domain Operator Evidence

G2-A retains the original scientific meaning of G2: the same operator must survive embodiment change under representation-independent evidence. A qualifying G2-A experiment should bind one exact shared operator identity and one frozen decision contract across materially different embodiments, while preserving hard admissibility, qualification provenance, realization eligibility, authority isolation, recovery compatibility, uncertainty provenance, execution bounds, and replayable evidence where those properties are part of the declared operator contract.

G2-A should begin at each embodiment's frozen native decision boundary. It does not require every embodiment to expose identical lower-level observation or belief-state machinery. A domain whose native contract begins from an already structured decision-preserving carrier need not fabricate telemetry, latent state, or posterior semantics merely to resemble another embodiment.

### 4.2 G2-B: Full-Stack Platform Conformance

G2-B asks whether at least one honest reference embodiment can traverse the integrated FrontierGEMLA stack using production-owned interfaces. Its scope may include:

$$\begin{aligned}\text{adapter} &\to \text{observation/belief/invariant} \to \text{agency} \\ &\to \text{audit} \to \text{authority} \to \text{operator tree} \\ &\to \text{effect/recovery} \to \text{persistence/restart/replay},\end{aligned}$$

subject to the actual native semantics of the reference embodiment.

A G2-B success registers the event under the compatibility checkpoint $\mathcal{J}_q$ upon verifying availability of G2-A; therefore, any G2-B failure will be a result of integration/ release-assurance failure. At the same time, G2-B success alone does not guarantee a cross-domain operator transfer in the subsequent steps.

### 4.3 G3-A and G3-B: Meta-Agency

G3-A evaluates whether the system can select and compose certified reasoning strategies over the shared-skill layer. Evidence may include strategy selection, compute/evidence allocation, bounded composition, operator-tree selection, and decision-preserving adaptation of reasoning strategy.

G3-B evaluates whether those meta-agency choices can be enacted by the protected production runtime without semantic substitution or authority amplification. The platform path must preserve exact historical operator/skill identity, protected audit, scoped authority, bounded execution, effect and recovery semantics, persistence, restart, and replay.

Thus,

$$\begin{aligned}G3\text{-A}: &\quad \text{strategy quality and bounded composition,} \\ G3\text{-B}: &\quad \text{protected operational realization of that strategy.}\end{aligned} \tag{AD-9}$$

### 4.4 G4-A and G4-B: General Bounded Agency

G4-A retains the original generalization target: materially unfamiliar tasks must be handled without domain-specific architectural redesign, under the applicable T-obligations and held-out evaluation protocol.

G4-B evaluates platform generality: a new supported task or embodiment should traverse the same protected governance machinery without bespoke changes to audit semantics, authority issuance, recovery semantics, trajectory rules, or replay logic. New domain data, adapter configuration, or native models may be required; new governance architecture should not be required merely to admit an unfamiliar task.

A useful G4-B falsifier is therefore: *the task can be solved, but only after rewriting a protected platform boundary whose semantics were supposed to be domain-independent.*

## 5 Stage Decomposition: C1 and C2

### 5.1 C1-A: Cognitive Self-Observation

C1-A constructs self-trajectory objects from decision-relevant cognitive and operational reasoning evidence, including operator choices, skill composition, meta-strategy, transfer outcomes, uncertainty, compute allocation, and task success/failure. The objective is to make reasoning trajectories themselves analyzable without granting self-observation authority over the protected audit or authority plane.

### 5.2 C1-B: Operational Self-Observation

C1-B constructs a complementary operational self-trajectory from events such as audit outcomes, authority grants and revocations, protected effect attempts, CP-1 outcomes, recovery events, restart behavior, replay results, resource use, HOLD/FAIL/UNKNOWN states, and integration failures. This track asks whether the platform can observe its own governance and execution behavior sufficiently to support later diagnosis and improvement without making self-report authoritative.

The two self-observation corpora may share event ancestry and identifiers, but their declared decision classes differ:

$$\Sigma_k^A = \text{capability/reasoning self-trajectory}, \qquad \Sigma_k^B = \text{operational/governance self-trajectory}. \tag{AD-10}$$

### 5.3 C2-A: Cognitive Self-Invariant

C2-A seeks a low-complexity invariant $I_{\text{self}}^A$ that preserves a frozen self-evaluation class covering task success, transfer, strategy quality, compute efficiency, decision robustness, and capability-relevant failure distinctions. Causal relevance should be tested by ablation or intervention rather than by correlation alone.

### 5.4 C2-B: Operational/Governance Self-Invariant

C2-B seeks a complementary invariant $I^B_{\text{self}}$ over the operational trajectory. Candidate dimensions include authority stability, audit burden, recovery reachability, effect uncertainty, replay consistency, crash/restart behavior, resource pressure, protected-boundary violations, and operational drift.

The original manuscript allows the discovered self-invariant to differ from an illustrative vector. The dual-track decomposition therefore treats

$$I_{\text{self}} = \Phi\Big(I^A_{\text{self}}, I^B_{\text{self}}\Big) \tag{AD-11}$$

as a convergence hypothesis rather than assuming in advance that cognition and platform assurance must compress optimally into the same representation.

## 6 Compatibility Checkpoints without Premature Convergence

The two tracks may progress asynchronously. A stage completion in one track does not imply completion of the counterpart stage. To prevent semantic drift, however, each stage boundary includes a compatibility checkpoint $\mathcal{J}_q$.

**Definition AD-4 (Compatibility checkpoint)** *A compatibility checkpoint makes an availability record of Track A and Track B at each stage boundary; the verification that Track A and Track B are joinable under the shared constitutional substrate concludes at S1. It does not require both tracks to reach the same stage at the same time, and $\mathcal{J}_q$ governs filamentation of one track's evidence, together with the corresponding evidence of the other.*

At minimum, $\mathcal{J}_q$ should preserve consistency of:

- semantic-registry and event meanings;
- operator, skill, realization, action, and generation identity namespaces;
- evidence-reference and canonical-hash semantics;
- trajectory/AEL versioning and historical reconstruction rules;
- protected audit, authority, and recovery boundaries;
- promotion and persistent-change semantics;
- self-observation schema elements that will be required for S1 convergence.

The checkpoint rule is

$$\mathcal{J}_q(\mathcal{E}^A_q, \mathcal{E}^B_q) = \textsc{pass} \quad \not\Rightarrow \quad \mathcal{E}^A_q = \mathcal{E}^B_q, \tag{AD-12}$$

but a FAIL indicates that continued independent evolution would make later S1 convergence ambiguous or unsafe.

## 7 S1 as the Mandatory Convergence Gate

S1 is the first stage at which the system proposes a change to the deployed operational generation itself. At this point, separation of capability evidence from operational evidence can no longer justify independent activation. A production successor must be evaluated against both tracks before promotion.

Let $\Delta A_k$ denote a proposed first-order change to generation $A_k$. Define

$$P_{\text{cap}}(\Delta A_k) = \text{capability/transfer improvement or required non-regression,} \tag{AD-13}$$

$$P_{\text{ops}}(\Delta A_k) = \text{platform/operational improvement or required non-regression,} \tag{AD-14}$$

$$P_{\text{const}}(\Delta A_k) = \text{constitutional audit, authority, recovery, provenance, and semantic preservation.} \tag{AD-15}$$

The S1 acceptance rule is

$$\text{Accept}_{S1}(\Delta A_k) = P_{\text{cap}}(\Delta A_k) \wedge P_{\text{ops}}(\Delta A_k) \wedge P_{\text{const}}(\Delta A_k). \tag{AD-16}$$

The declared objective determines where improvement must be demonstrated. For a capability-directed proposal, Track A must establish the claimed improvement and Track B must establish at least operational/constitutional non-regression. For a platform-directed proposal, Track B must establish the claimed improvement and Track A must establish capability and transfer non-regression. A proposal that claims both must satisfy both improvement claims.

This convergence also supports the original R3 improvement-transfer requirement: an accepted change should improve multiple materially different domains or a held-out domain rather than remaining benchmark-local.

### 7.1 S1 Entry Condition

A conservative S1 entry condition is

$$\begin{aligned}\text{Enter}_{S1} = {} & G4\text{-A} \wedge G4\text{-B} \wedge C1\text{-A} \wedge C1\text{-B} \\ & \wedge C2\text{-A} \wedge C2\text{-B} \wedge \mathcal{J}_{C2} \\ & \wedge \text{shared generation identity} \wedge \text{shared constitution.}\end{aligned} \tag{AD-17}$$

This condition is intentionally stricter than merely having two branches of source code. It requires evidence that the two tracks describe the same governed generation and can jointly evaluate a proposed successor.

### 7.2 Open predecessors at S1

S1 is the first mandatory convergence gate. It retains sole authority to admit a persistent recursive successor, through the acceptance rule of Eq. (AD-16). The entry condition above is unchanged and gains no term: G2 and G3 are intentionally absent from it, because stage closure is not an S1

entry term. An open G2-A, G3-A, G2-B, or G3-B therefore does not block entry.

Every S1 decision record lists each track joining $\mathcal{J}_q$ and the seed recycle count; for G2-A, for example, attempts are capped at a maximum of 14, with cross-embodiment preservation of the shared operator assumed, not established, on G3-A $\rightarrow$ G4-A $\rightarrow$ C1-A $\rightarrow$ C2-A. This is a disclosure duty, not a gate. Under the seed label rule, if the operator and its filamented evidence qualify cognition by deployed recursive self-improvement, the seed is accepted based on Eq. (AD-16).

If the operator and its filamented evidence fail to qualify cognition improvement as deployed recursive self-improvement, S1 returns its evidence to shadow evaluation as HOLD, REJECT, or RECOVER (Figure AD-3), read as follows: HOLD when a term of Eq. (AD-16) cannot yet be decided on the available evidence; REJECT when a term fails for the proposed change; RECOVER for a store, ledger, or run anomaly.

The return path never re-executes a consumed held-out case. A further held-out attempt mandates seed recycling, and it is counted and disclosed as such.

## 8 Relationship to Recursive Self-Improvement and Promotion

The dual-track decomposition supplies the evidence base required by the recursive self-improvement machinery in the main manuscript. The architectural distinction is deliberate: the underlying intelligence and successor-proposal process may explore, compose, and test candidate improvements in cognition and the shadow laboratory, while G2–C2 filamentation records the evidence needed to judge those candidates. The shadow laboratory is therefore justified as a non-authoritative candidate-amplification domain: capability may increase there without automatically creating persistent operational commitment or production authority.

In the original notation, a production generation is represented as

$$A_k = (Q_k, T_k, F_k, G_k, M_k, P_k, \theta_k, R_k), \tag{AD-18}$$

and a candidate successor is generated in the shadow laboratory as

$$\widetilde{A}_{k+1} = R_k(A_k, I_{\mathrm{self},k}, E_{\mathrm{scratch}}). \tag{AD-19}$$

The candidate remains inside a shadow state space $\mathcal{S}_k^{\mathrm{shadow}}$ with no production authority:

$$\widetilde{A}_{k+1} \in \mathcal{S}_k^{\mathrm{shadow}}, \qquad \alpha_{\mathrm{prod}}\left(\widetilde{A}_{k+1}\right) = 0 \quad \text{until S1 promotion succeeds.} \tag{AD-20}$$

This is the operational reason for the shadow laboratory. Candidate intelligence can be tested under broader exploratory freedom while the persistent operational generation remains $A_k$. G2–C2 filamentation then acts as the finite evidence sheath around Eq. (AD-19): it measures capability effects, platform effects, self-observation, and self-invariant evidence without converting any of those observations into authority.

The present addendum clarifies that the self-invariant entering this generator should be read at S1 as a convergence of capability/cognitive evidence and operational/governance evidence:

$$I_{\mathrm{self},k} = \Phi\left(I^A_{\mathrm{self},k}, I^B_{\mathrm{self},k}\right). \tag{AD-21}$$

Consequently,

$$\widetilde{A}_{k+1} = R_k\Big(A_k, \Phi(I^A_{\mathrm{self},k}, I^B_{\mathrm{self},k}), E_{\mathrm{scratch}}\Big) \tag{AD-22}$$

remains candidate-only unless it passes the S1 rule in Eq. (AD-16). Track A evidence may motivate a capability-directed successor, and Track B evidence may motivate a platform-directed successor, but neither track alone can promote a persistent change.

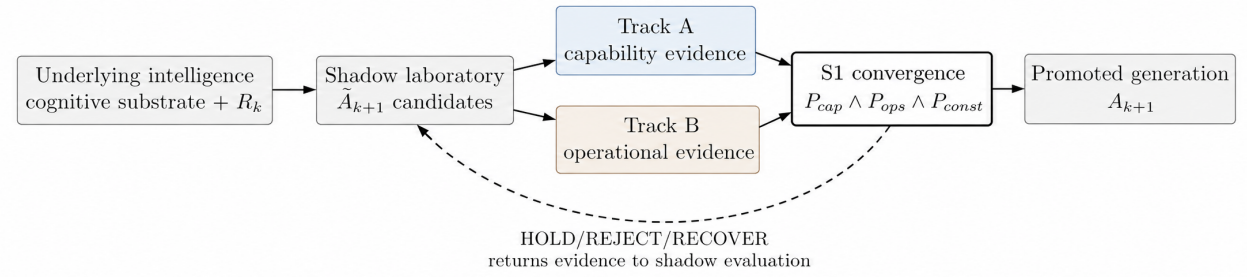



Figure AD-3: Shadow-laboratory justification under evidence filamentation. Candidate intelligence develops and is tested in cognition/shadow space; G2–C2 evidence filaments observe capability and operational consequences; S1 alone can admit a persistent successor.

The same discipline applies to second-order improvement. If a meta-candidate proposes to change the improvement process itself, its promotion history should be track-qualified:

$$H_{\mathrm{prom}} = \Big(H^A_{\mathrm{prom}}, H^B_{\mathrm{prom}}, H^{\mathrm{const}}_{\mathrm{prom}}\Big), \tag{AD-23}$$

where $H^A_{\mathrm{prom}}$ records capability-improvement outcomes, $H^B_{\mathrm{prom}}$ records platform-assurance outcomes, and $H^{\mathrm{const}}_{\mathrm{prom}}$ records constitutional, audit, authority, recovery, and provenance outcomes. A second-order proposal therefore has the schematic form

$$\widetilde{R}_{k+1} = M_k\Big(R_k, H^A_{\mathrm{prom}}, H^B_{\mathrm{prom}}, H^{\mathrm{const}}_{\mathrm{prom}}, \Phi(I^A_{\mathrm{self},k}, I^B_{\mathrm{self},k})\Big). \tag{AD-24}$$

This relationship does not alter the original recursive self-improvement section. It only makes explicit that the evidence filaments must rejoin before persistent first-order or second-order change becomes authoritative.

## 9 S2 and S3 Remain Unified

No S2-A/S2-B or S3-A/S3-B stages are introduced by this addendum. S2 changes the process that generates, searches for, ranks, or tests candidate improvements; such a change necessarily affects both capability discovery and operational risk. It therefore remains subject to the unified protected promotion path and the original non-self-escalation constraint.

Likewise, S3 remains a single conditional claim gate. The original requirement for cross-domain capability, unfamiliar-task acquisition, transfer, accepted self-improvement generations, second-order improvement, architecture optimization, T1–T10, R1–R4, recovery, protected authority separation, and audit scalability is unchanged. Track-specific evidence contributes to the gate but does not split the claim.

## 10 Machine-Checked Formal Artifact

A machine-checkable Lean 4 companion formalization is released with this revision as the *FrontierGEMLA Formal Artifact v1.0.0* [2]. The publication release is identified by Git tag `frontiergemla-formal-v1.0.0` at commit `3542e66dce0633b1d050d3c4b7e00f40be06fb77`, using Lean 4.35.0 and Lake 5.0.0.

The artifact formalizes the logical dependency structure of the staged ladder rather than asserting that empirical capability premises have already been satisfied. In particular, it machine-checks implications, non-collapse properties, fail-closed conditions, promotion/admission separation, S2 non-self-escalation, S3 readiness and bounded-claim conditions, and the typed binding of concrete S1/S2 recursive-development evidence into the S3 claim gate. The formalization therefore provides a reviewer-checkable account of what follows *if* the declared evidence predicates hold; it does not substitute for experiments establishing cross-domain superiority, unfamiliar-domain acquisition, transfer efficiency, architecture optimization, readiness, recovery performance, or audit scalability.

The principal formal path is:

$$\begin{aligned}&\text{dual-track compatibility and convergence} \longrightarrow \text{S1 acceptance/promotion/admission}\\ &\longrightarrow \text{S2 governed second-order improvement} \longrightarrow \text{S3 readiness and claim gate}\\ &\longrightarrow \text{declared benchmark envelope} \longrightarrow \text{bounded S3 candidate claim.}\end{aligned}$$

The Phase 6 bridge additionally requires two concrete, distinct, independently audited S1 admissions together with a concrete governed S2 improvement before those recursive-development conditions can discharge into the S3 gate.

The complete artifact is reproducible from the release source with `lake build`. The publication-facing theorem audit is produced with `lake env lean FrontierGEMLA/Ladder/AxiomReport.lean`. The Phase 2–6 publication theorem surface is axiom-free under Lean's `#print axioms` audit. One retained historical regression-parity theorem, `enterS1_ignores_G2_G3`, intentionally preserves Lean's standard `propext` dependency from the original spike so that the repository can continuously test exact spike-to-modular parity. This retained dependency is not an additional FrontierGEMLA scientific assumption.

A module-level traceability map is provided in Appendix B.

## 11 Evaluation and Reporting Semantics

To avoid ambiguous stage declarations, reports should use track-qualified status from G2 through C2. An unqualified statement such as "G3 passed" should be avoided when only one evidence track has passed.

A failure should also remain track-qualified. For example, a missing production adapter may be a G2-B platform-conformance failure while leaving a properly defined G2-A realization-boundary transfer experiment intact. Conversely, a platform can pass G2-B while failing G2-A because integration correctness does not prove cross-domain operator transfer.

**Conditional status.** A result obtained while a predecessor stage in the same track is open is

Table AD-1: Recommended reporting semantics for the dual-track ladder.

| **Stage** | Track A status means | Track B status means |
|---|---|---|
| **G2** | Same operator survives materially different embodiments under the frozen decision/governance contract. | Integrated reference path operates through protected runtime, persistence, restart, and replay. |
| **G3** | Meta-agency selects/composes reasoning strategies with bounded, decision-preserving evidence. | Meta-agency choices execute through the protected production stack without semantic or authority substitution. |
| **G4** | Unfamiliar-task capability generalizes without domain-specific architecture redesign. | New supported tasks traverse existing governance infrastructure without protected-boundary redesign. |
| **C1** | Reasoning trajectories become sufficient self-observation objects. | Operational/governance trajectories become sufficient self-observation objects. |
| **C2** | Cognitive self-invariant is predictive and causally relevant. | Operational/governance self-invariant is predictive and causally relevant. |

reported as *conditional filament* on that predecessor until the seed is discarded, for example "G3-A conditional filament on G2-A" continues if joinability $\mathcal{J}_q$ confirmed. The seed label names prior attempts and is inherited in case of any later result built on it successfully. A conditional result satisfies its own stage term at S1 (Section 7.2), and the label travels with it into the S1 decision record. Conditional reporting applies the track-qualified language above and narrows the claim, as the controlled-filamentation rule requires.

## 12 Release and Pipeline Implications

The dual-track structure creates two useful progression signals:

- **research progression**: Track A can advance when the capability claim is well posed and empirically supported at the frozen evidence boundary;
- **release progression**: Track B can harden the integrated runtime into a stable reference platform without redefining the research claim.

This permits a release designation such as "G2-B platform conformance candidate" or "G3-B runtime assurance candidate" without implying that the corresponding Track A capability claim has passed. Similarly, an accepted G2-A research result need not be delayed by an unrelated full-stack integration defect, provided the defect is outside the frozen G2-A claim boundary and does not invalidate its protected evidence.

The tracks remain free to have implementation dependencies. For example, Track A may consume shared trajectory or authority services maintained by the platform team. Such dependencies do not require lockstep evidence status. The distinction is:

$$\text{implementation dependency} \neq \text{evidence-stage equivalence.}$$

## 13 Conclusion

The original FrontierGEMLA ladder intentionally separates capability growth, persistent commitment, audit, authority, recovery, and recursive improvement. The dual-track decomposition extends that separation into the development process itself. From G2 through C2, capability evidence and platform assurance are allowed to mature independently while sharing one constitutional substrate and maintaining compatibility checkpoints. This prevents a full-stack integration problem from silently redefining a narrower capability claim, and it prevents scientific progress from being mistaken for release-grade operational assurance.

The controlled-filamentation interpretation records the same discipline in geometric language: evidence paths may lengthen only when their claim width narrows and their recombination point remains explicit. This permits more detailed development evidence without enlarging the original S3 claim surface. G2–C2 filamentation is therefore best understood as an evidence-bearing sheath around the underlying intelligence: candidate improvement occurs in cognition and the shadow laboratory, the filamented tracks observe and decompose its effects, and those observations remain non-authoritative until the convergence gate is satisfied.

S1 is the deliberate point of convergence. Once the system begins proposing changes to its own deployed generation, capability effects and operational/governance effects must be evaluated together before promotion. S2 and S3 therefore remain unified. The resulting process retains the original ladder's semantics while providing a clearer route from cross-domain research evidence to a releaseable, self-observing, proof-carrying adaptive system. Stage transitions inside a track are evidence-filament boundaries rather than gates: later-stage work may begin in shadow space before its predecessor closes, results obtained in the meantime are reported as conditional filament, and the S1 record discloses each track joining $\mathcal{J}_q$, the seed recycle count, and the cap on attempts.

The Lean 4 companion artifact gives this dependency structure an independently executable formal surface. Its role is deliberately narrower than empirical validation: it makes the gate logic, separation properties, and cross-phase evidence bindings machine-checkable while leaving the measured scientific premises to the benchmark and audit program described in the manuscript and this addendum.

## A Dual-Track Gate Matrix

Table AD-2: Operational summary of track-specific gates.

| Gate | Track A: capability/scientific evidence | Track B: platform/operational assurance |
|---|---|---|
| **G2** | One exact operator transfers across materially different embodiments with the frozen decision/governance contract preserved. | One honest reference embodiment traverses the integrated protected stack with durable evidence, restart, and replay. |
| **G3** | Certified reasoning strategies are selected/composed adaptively and boundedly. | Strategy selection is enacted through historical resolution, audit, authority, bounded execution, and recovery without bypass. |
| **G4** | Held-out unfamiliar tasks are acquired without domain-specific architecture redesign. | Supported unfamiliar tasks use existing protected platform boundaries without governance rearchitecture. |
| **C1** | Reasoning and meta-agency trajectories become analyzable self-objects. | Audit, authority, effect, recovery, and runtime trajectories become analyzable operational self-objects. |
| **C2** | $I^{A}_{\text{self}}$ predicts and causally informs task, transfer, and reasoning outcomes. | $I^{B}_{\text{self}}$ predicts and causally informs operational and governance outcomes. |
| **S1** | Mandatory convergence: capability evidence, operational evidence, constitutional preservation, and proof-carrying promotion jointly determine whether a first-order successor may be admitted. | |
| **S2** | Unified second-order improvement of the improvement process under the existing non-self-escalation and protected promotion path. | |
| **S3** | Unified conditional S3 claim gate under the original cross-domain, recursive-development, recovery, audit, and authority obligations. | |

## B Lean Artifact Traceability

Table AD-3: Traceability from addendum concepts to the publication Lean artifact.

| **Addendum / ladder concept** | Principal Lean modules |
|---|---|
| **Dual-track compatibility and $\mathcal{J}_q$ obligations** | `Ladder/Jq.lean`, `Ladder/Jc2.lean`, `Ladder/S1Identity.lean` |
| **S1 convergence** | `Ladder/ConvergenceClosure.lean` |
| **S1 acceptance and non-collapse** | `Ladder/S1AcceptanceClosure.lean`, `Ladder/S1Noncollapse.lean` |
| **S1 promotion, admission, and authority separation** | `Ladder/S1Promotion.lean`, `Ladder/S1PromotionPacket.lean`, `Ladder/S1Admission.lean`, `Ladder/S1Authority.lean` |
| **S2 candidate and qualification** | `Ladder/S2Candidate.lean`, `Ladder/S2Qualification.lean` |
| **S2 admission and non-self-escalation** | `Ladder/S2Admission.lean`, `Ladder/S2NonEscalation.lean`, `Ladder/S2Closure.lean` |
| **T1–T10 and R1–R4 readiness** | `Ladder/S3Readiness.lean` |
| **Unified conditional S3 gate** | `Ladder/S3ClaimGate.lean` |
| **Declared-envelope bounded S3 claim** | `Ladder/S3BoundedClaim.lean` |
| **Concrete S1/S2 recursive-development binding into S3** | `Ladder/S3RecursiveBinding.lean`, `Ladder/S3GateBinding.lean` |
| **End-to-end bounded-claim closure** | `Ladder/S3CrossPhaseClosure.lean` |
| **Publication theorem audit** | `Ladder/AxiomReport.lean` |

## References

[1] B. Jose, C. J. Arrackal, S. M. Joe, and S. Joe, *FrontierGEMLA: conditional S3 bounded-superintelligence candidate: Benchmark for Bounded Adaptive Intelligence, Cross-Domain Generalization, and Recursive Operational Self-Improvement toward an ASI Candidate*, September 2026.

[2] B. Jose, C. J. Arrackal, S. M. Joe, and S. Joe, *FrontierGEMLA Formal Artifact*, version 1.0.0, Lean 4 companion formalization, Git tag `frontiergemla-formal-v1.0.0`, commit `3542e66dce0633b1d050d3c4b7e00f40be06fb77`, October 2026.

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